

Total Appointments

5000

Total No-Shows

2423

No-Show Rate

51.2%

Attended Appointments

2314

Cancelled Appointments

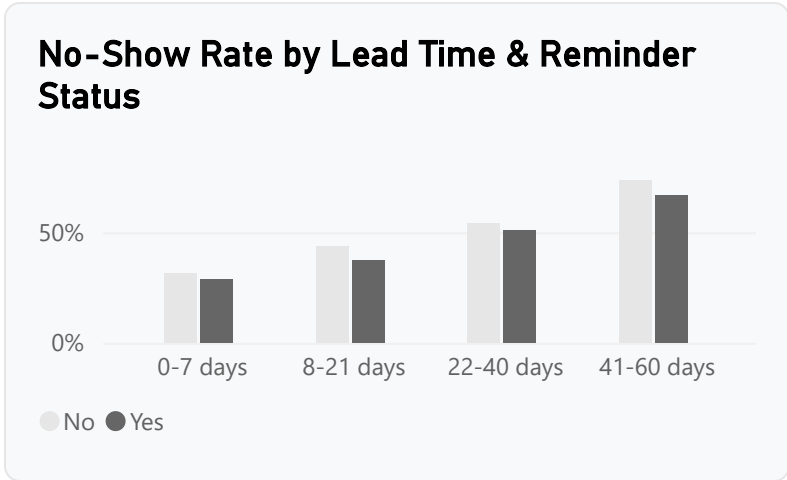
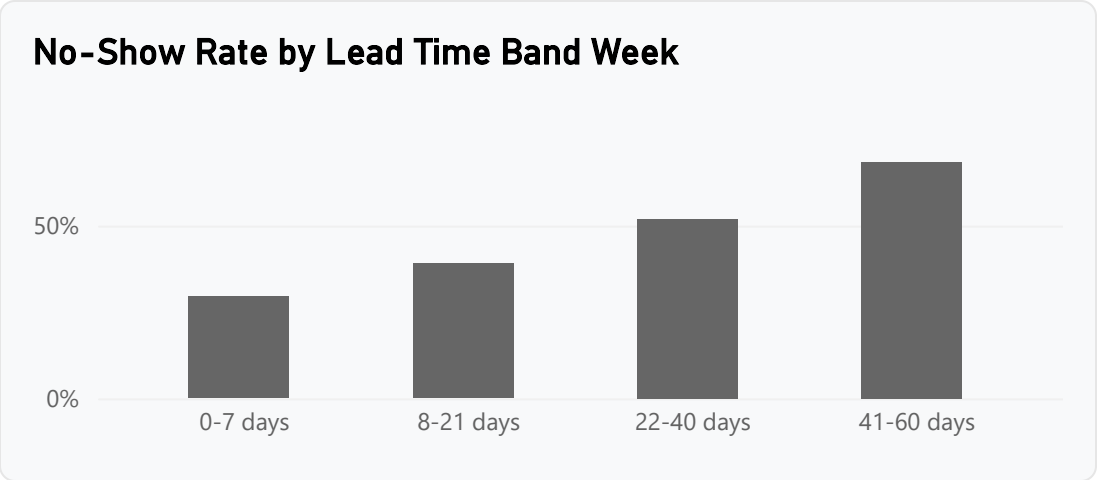
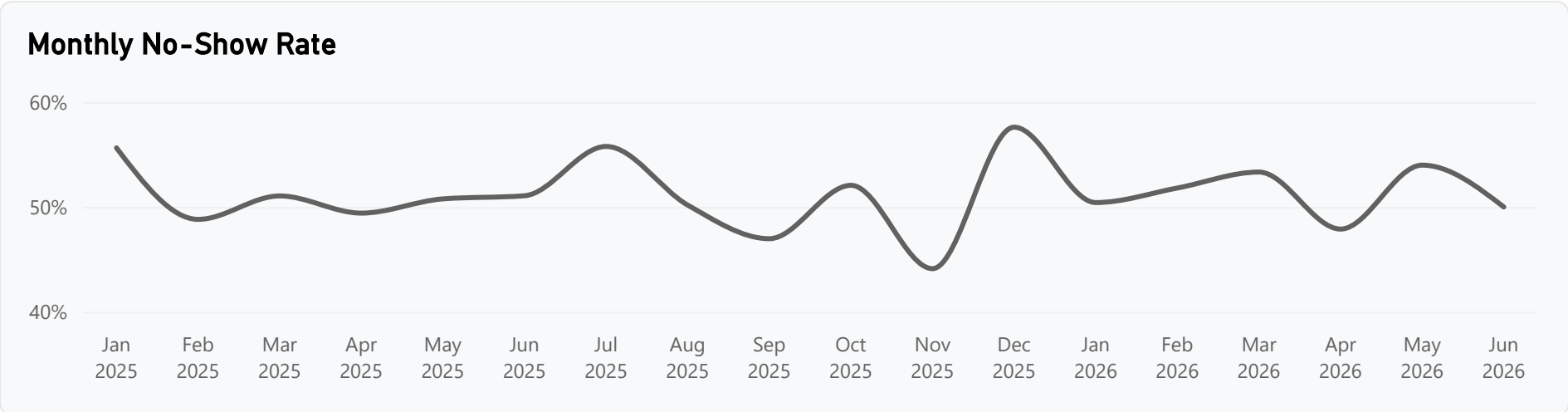
263

Cancellation Rate

5.26%

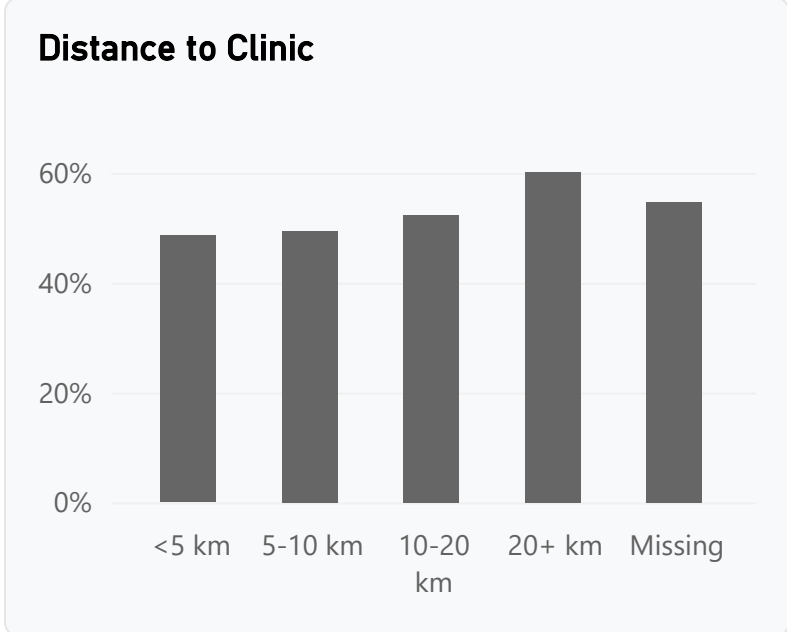
Eligible Appointments

4737



### No-Show Risk: Lead Time \* Previous History

| Previous No-Show History | 0-7 days | 8-21 days | 22-40 days | 41-60 days |
|--------------------------|----------|-----------|------------|------------|
| No Previous No-Show      | 22.8%    | 35.9%     | 48.2%      | 62.8%      |
| Previous No-Show         | 39.0%    | 43.8%     | 56.8%      | 75.7%      |
| Total                    | 29.5%    | 39.0%     | 51.8%      | 68.5%      |



Key Insights

Distance matters. Patients living 20+ km from the clinic miss their appointments 60.2% of the time. This geographic risk factor is a much stronger predictor of no-shows than patient age or the day of the week.

The no-show rate jumps to a critical 83.3% when long booking lead times (41–60 days) combine with a history of 2+ previous no-shows.

No-show rates rise substantially as booking lead time increases, reaching approximately 68.5% for appointments booked 41–60 days ahead.

Reminder-associated appointments recorded lower observed no-show rates within each lead-time group.

### Priority No-Show Segments

| Segment                           | No-Show Rate | Priority |
|-----------------------------------|--------------|----------|
| 41-60 days + 2+ previous no-shows | 83.3%        | Critical |
| 41-60 day lead time               | 68.5%        | High     |
| 2+ previous no-shows              | 63.5%        | High     |
| 20+ km from clinic                | 60.2%        | High     |
| Any previous no-show              | 57.8%        | High     |
| No reminder                       | 54.6%        | Moderate |
| Follow-up appointment             | 54.2%        | Moderate |

# Advanced Analysis & Data Science integration

Model ROC-AUC

68.9%

Model Recall

65.8%

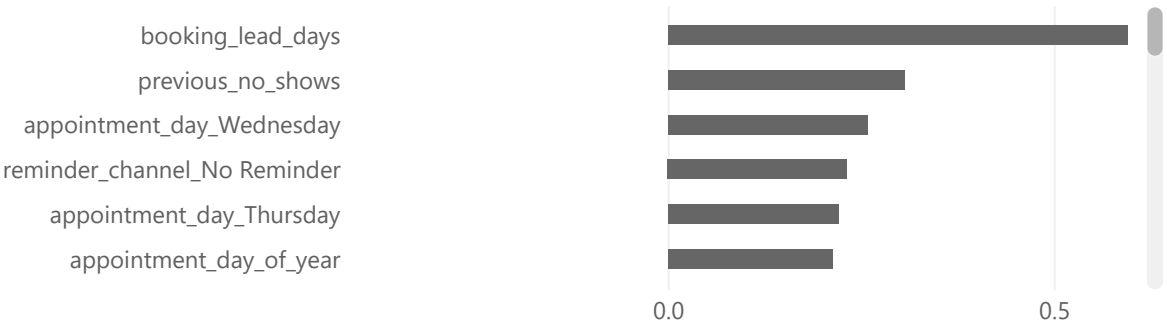
Model Precision

63.0%

Model F1

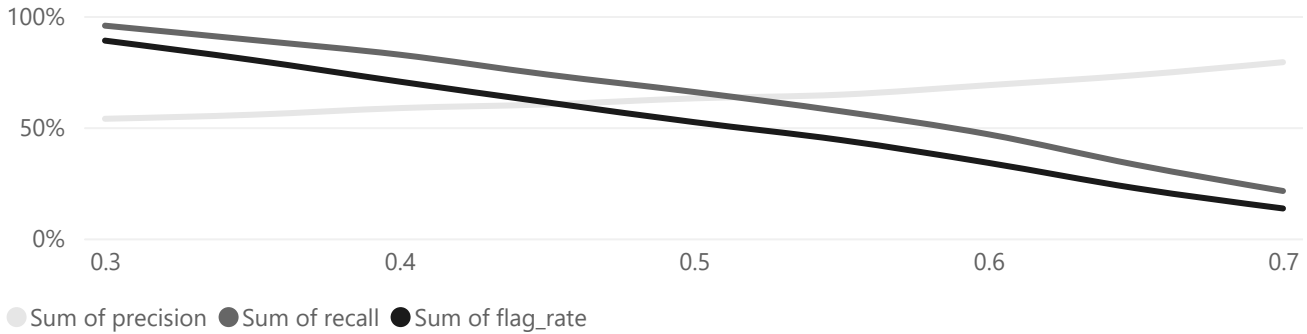
64.4%

## Logistic Regression Coefficients



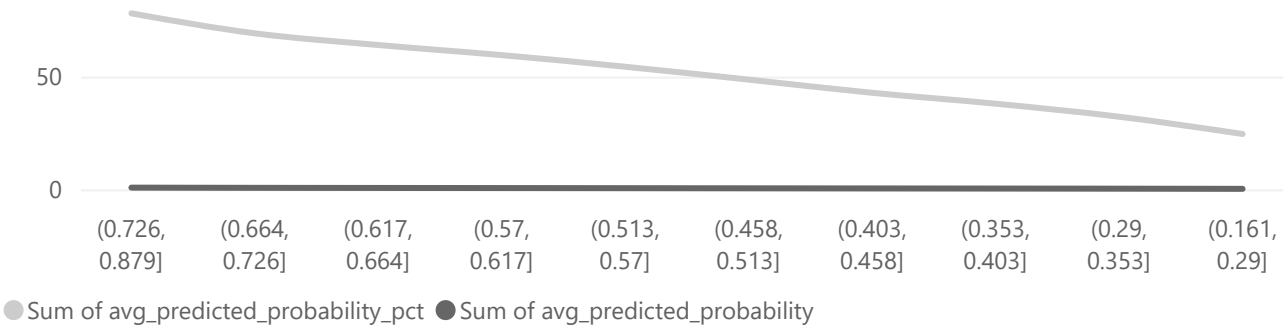
Positive coefficients are associated with higher predicted no-show probability; negative coefficients are associated with lower predicted probability, subject to model encoding/reference categories.

## Classification Threshold Trade-Off



Lower thresholds capture more potential no-shows but require more outreach. Higher thresholds target fewer appointments but miss more true no-shows. Final threshold selection requires clinic capacity and cost assumptions.

## Predicted Probability vs Actual No-Show Rate



Actual no-show rates rise from approximately 16.5% in the lowest score decile to 77.3% in the highest, indicating useful risk ranking.

## Analytics \* Data Science Evidence

| Analytics Evidence               | Model Evidence                   | Interpretation                       |
|----------------------------------|----------------------------------|--------------------------------------|
| Lead time: 29.5% → 68.5%         | Strong predictive contribution   | Strongest shared risk indicator      |
| Previous history: 57.8% vs 46.3% | Positive predictive contribution | Prioritise supportive outreach       |
| 20+ km: about 60.2%              | Distance contributes             | Access burden deserves attention     |
| No reminder: 54.6% vs 49.9%      | Reminder variables contribute    | Test reminder strategy prospectively |
| Follow-up: about 54.2%           | Modest predictive contribution   | Secondary / interacting factor       |

## Integration limitation

The current scoring file contains model/test-row indices rather than the original HC-xxxxx appointment IDs. It has therefore not been joined to the main appointment dataset. A corrected scoring export containing the original appointment and patient identifiers is required for Week 7 row-level error analysis.